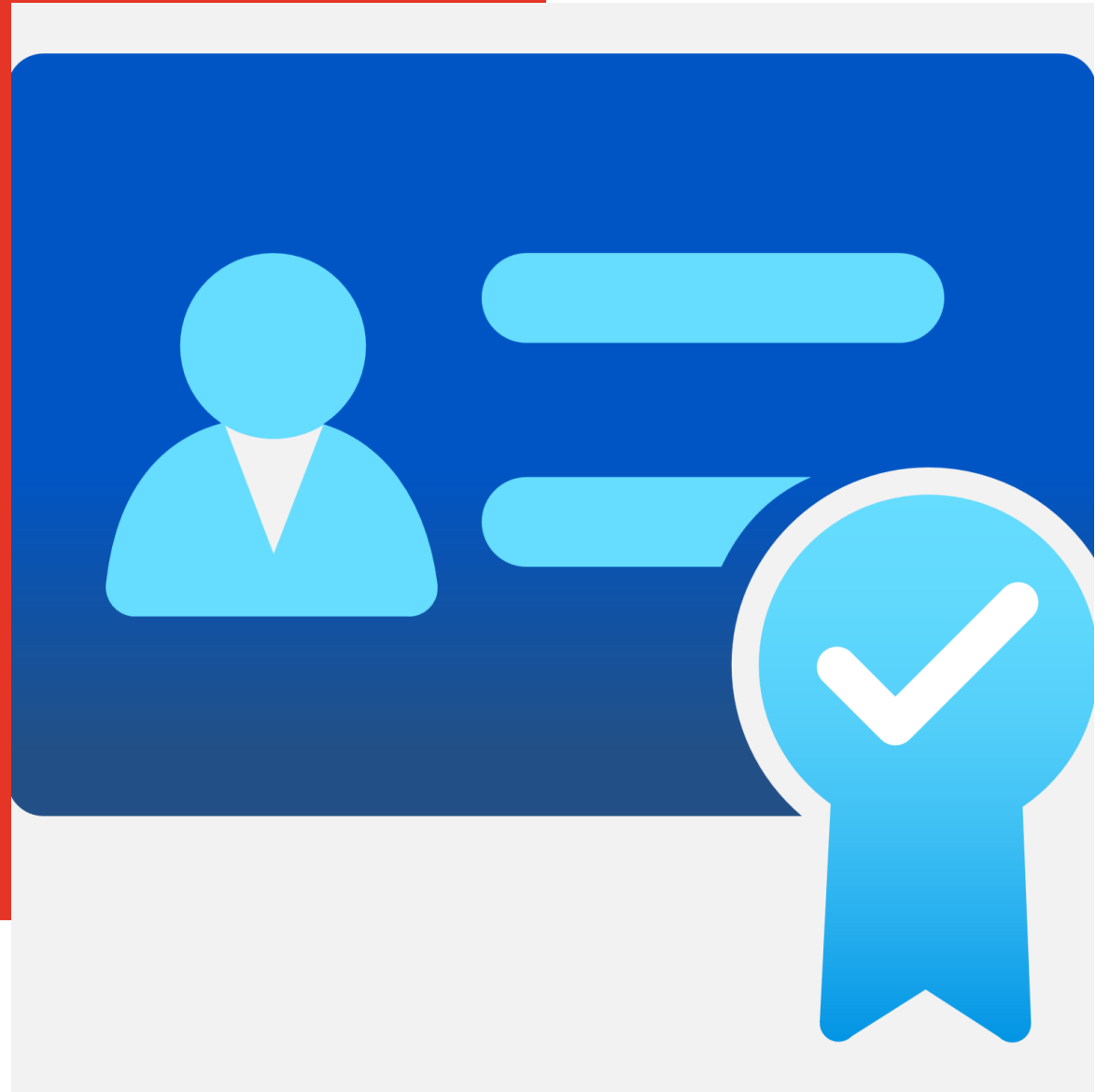


# Microsoft Entra Verified ID

Good, better, best option to secure issuance of Verifiable Credentials

Stefan Strube  
Solution architect



**Strukton**



# DANK AAN ONZE SPONSORS!



**R**ubicon



INSPARK

**valid**

**inforcer**



**INTERCEPT**  
MASTERS IN IT INFRA

**INNVOLVE**  
INN IT TOGETHER



3 maart Leusden

14 april Amstelveen

16 juni Eindhoven

Begin september n.t.b.

10 november Zwolle

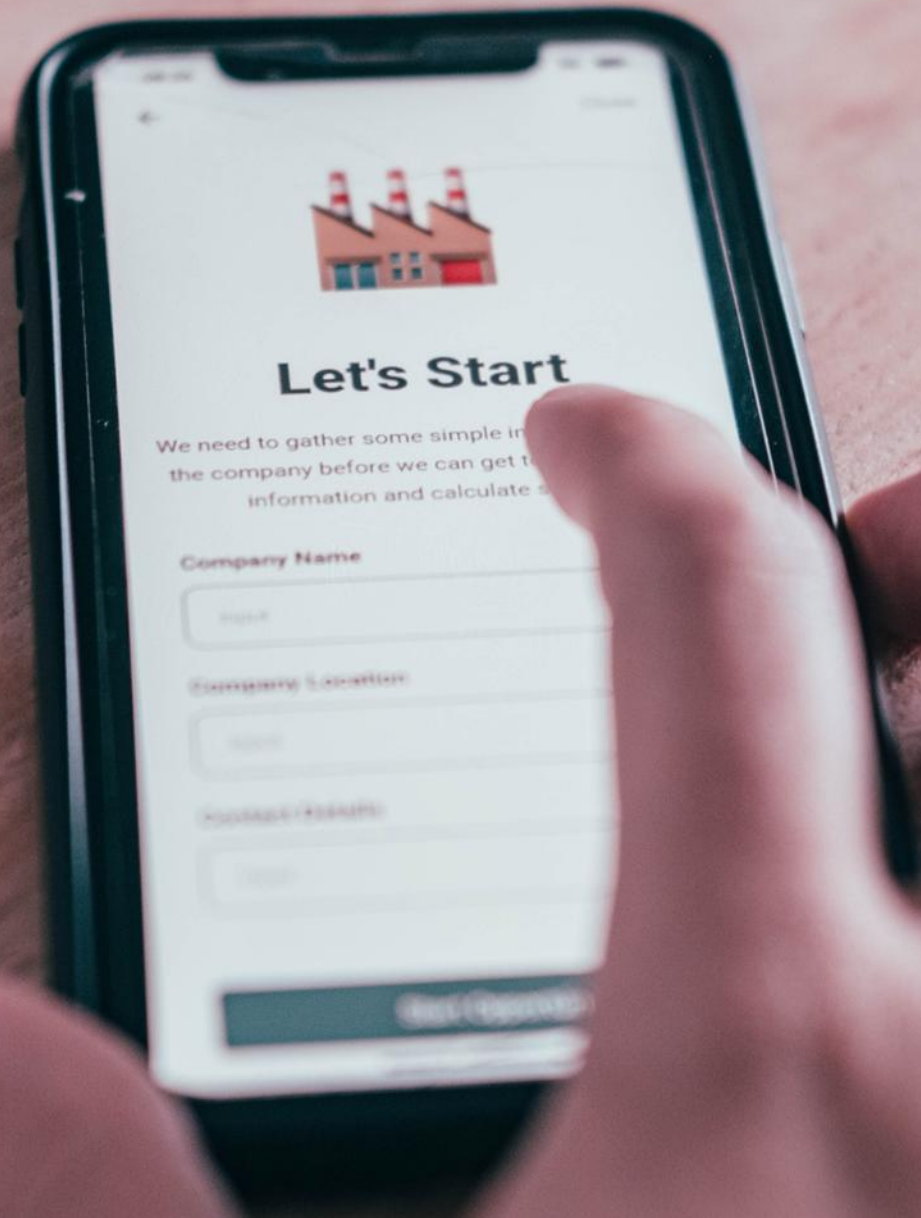
8 december Vught





# Join the demo

- Microsoft Forms



# Stefan Strube

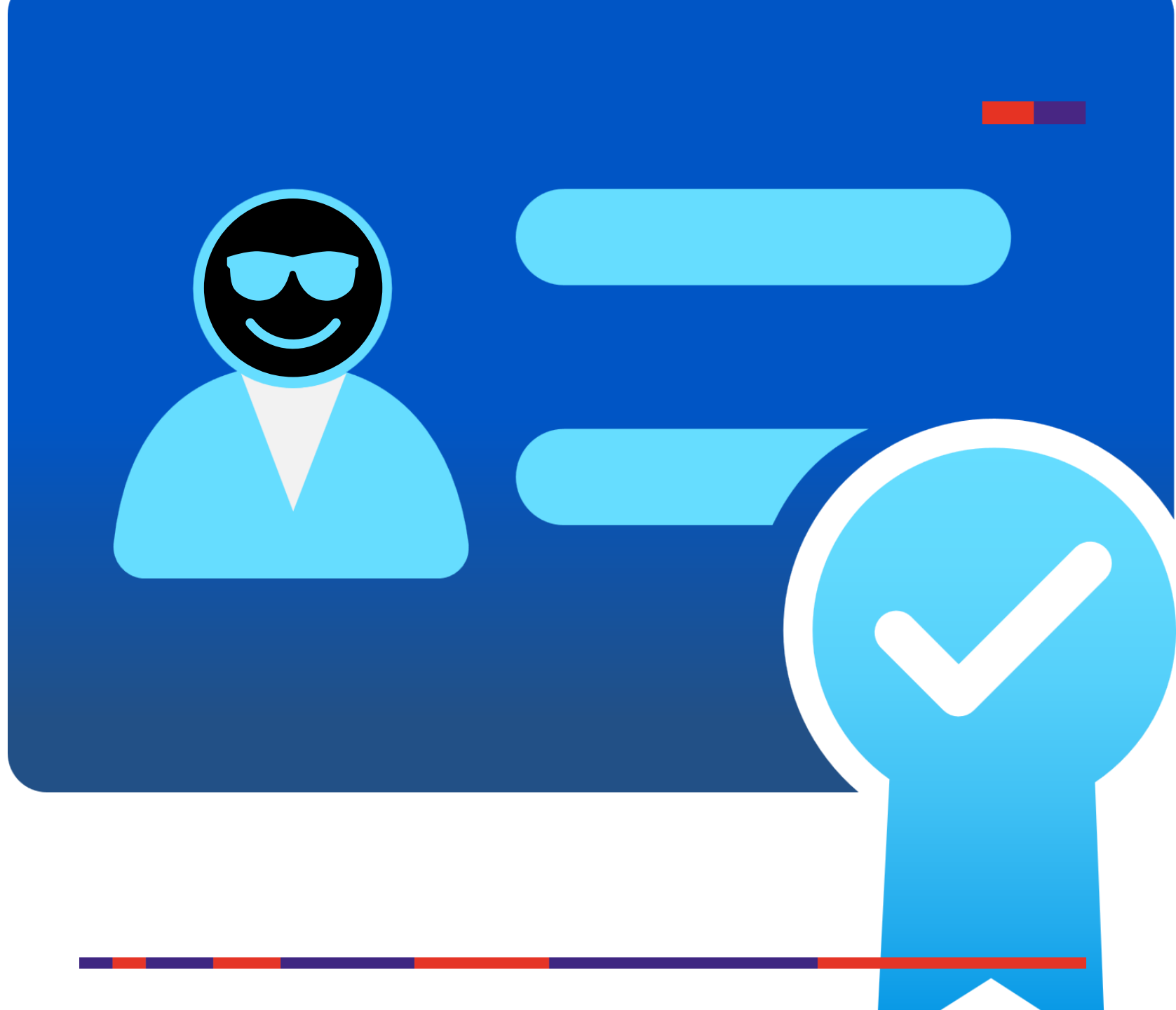
- Solution architect @ Strukton (NL)
- Microsoft Business Applications MVP
- Co-founder PowerAddictsNL user group



# Microsoft Entra Verified ID

Audience knowledge

- Heard of? 🙋
- Played with? 🙋
- Implemented? 🙋





# Own and control your digital identity with Microsoft Entra Verified ID

Anco Stuij | Rubicon B.V.

14 maart 2023







COMMUNITY DAY



Try it out yourself  
<https://aka.ms/diddemo>



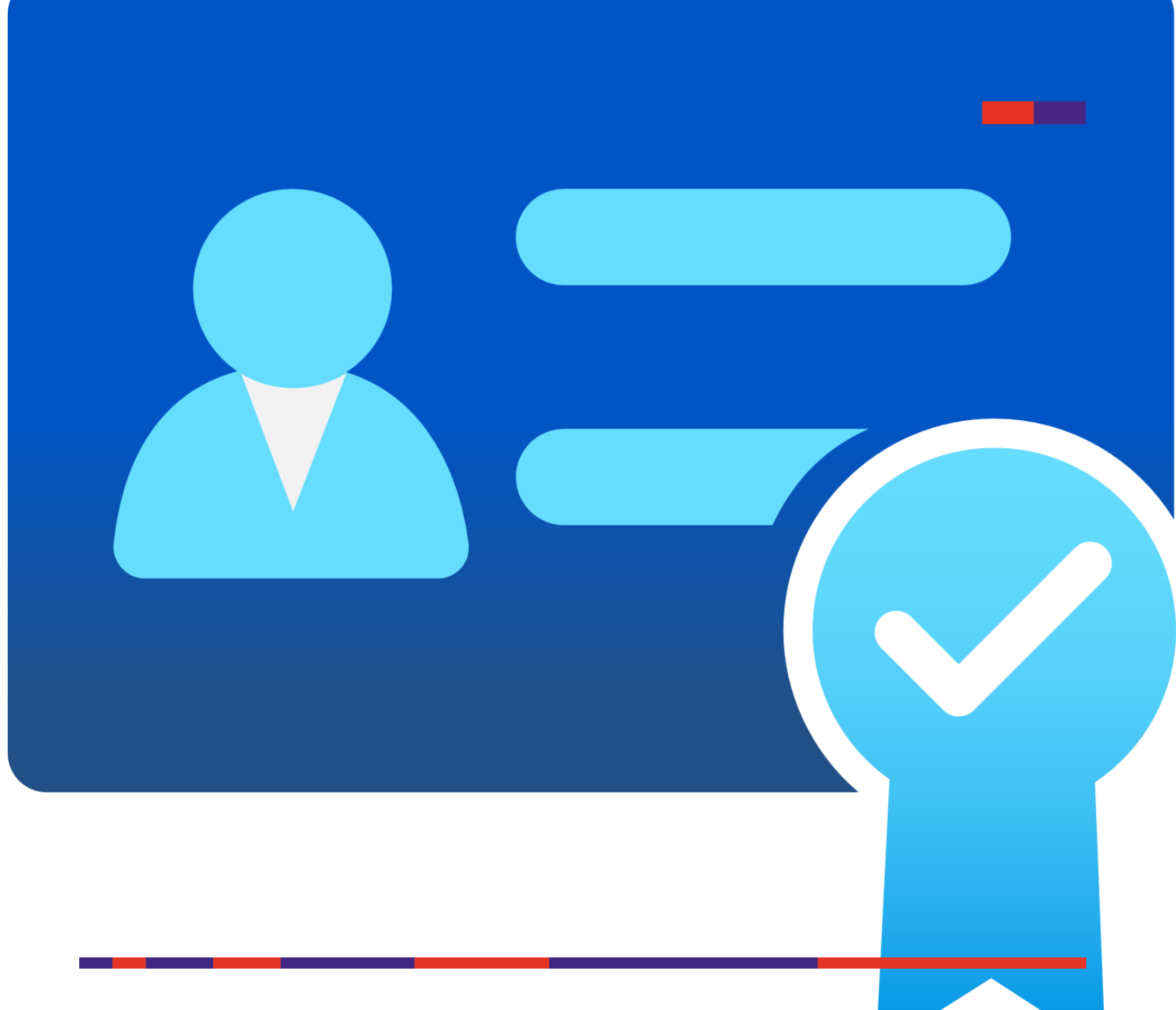
Build it yourself  
<https://aka.ms/ilovecats>



# Agenda

- Intro Microsoft Entra Verified ID
- How to start
- What's new
- Issue Verifiable Credentials
- Options to secure issuance
- Advanced scenarios

❖ Demos !



## Warning / Disclaimer !



Microsoft Entra Verified ID is  
not a primary identity / factor  
for authentication



# Intro Microsoft Entra Verified ID



Decentralized identity system

- W3C Decentralized Identifiers (DIDs)
- Trust System (shared network)
- DID User Agent / Wallet
- Microsoft Resolver (DID Lookup)
- Microsoft Entra Verified ID Service
- W3C Verifiable Credentials (VC)





# Intro Microsoft Entra Verified ID

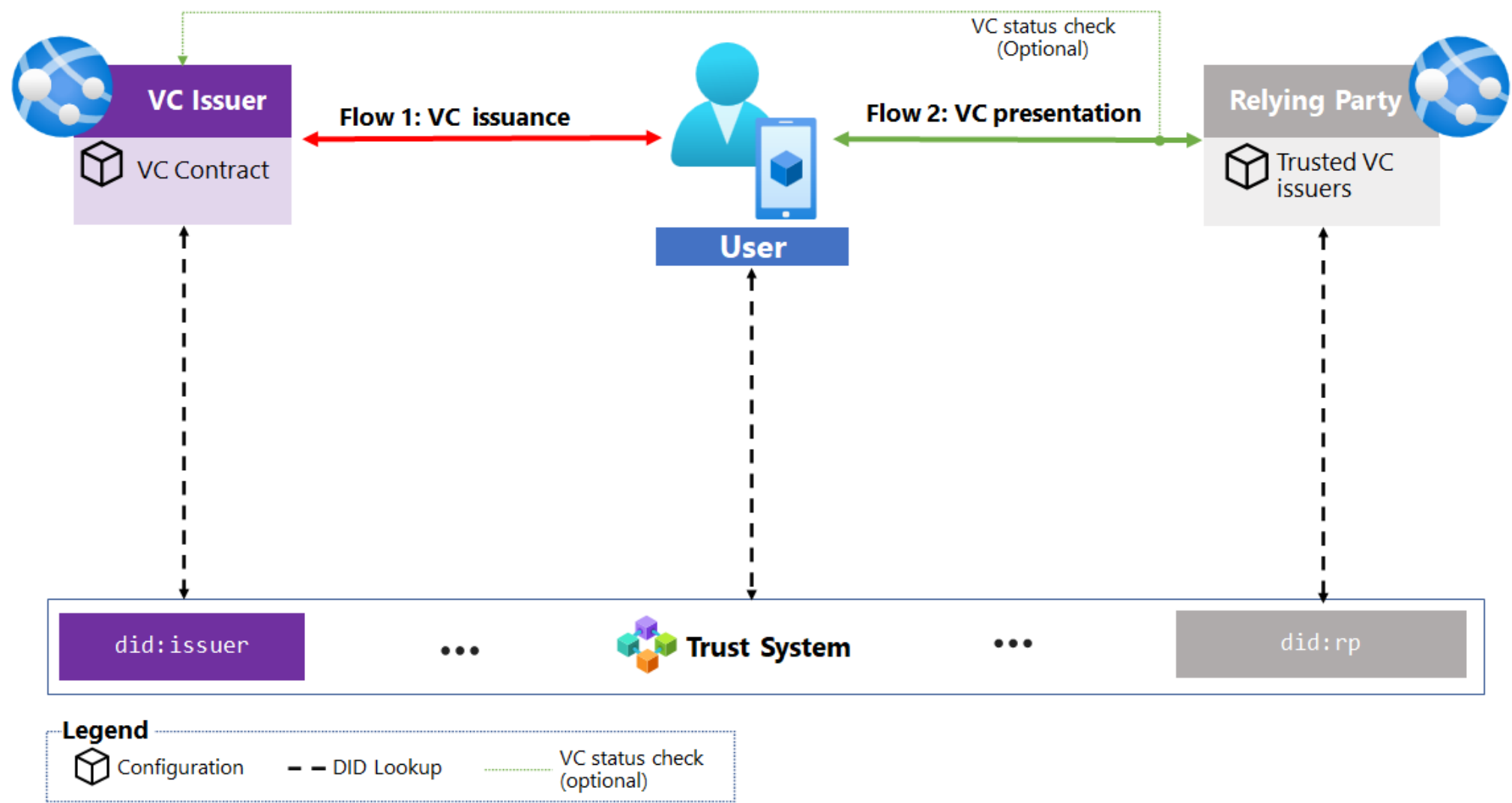


What are Verifiable Credentials?

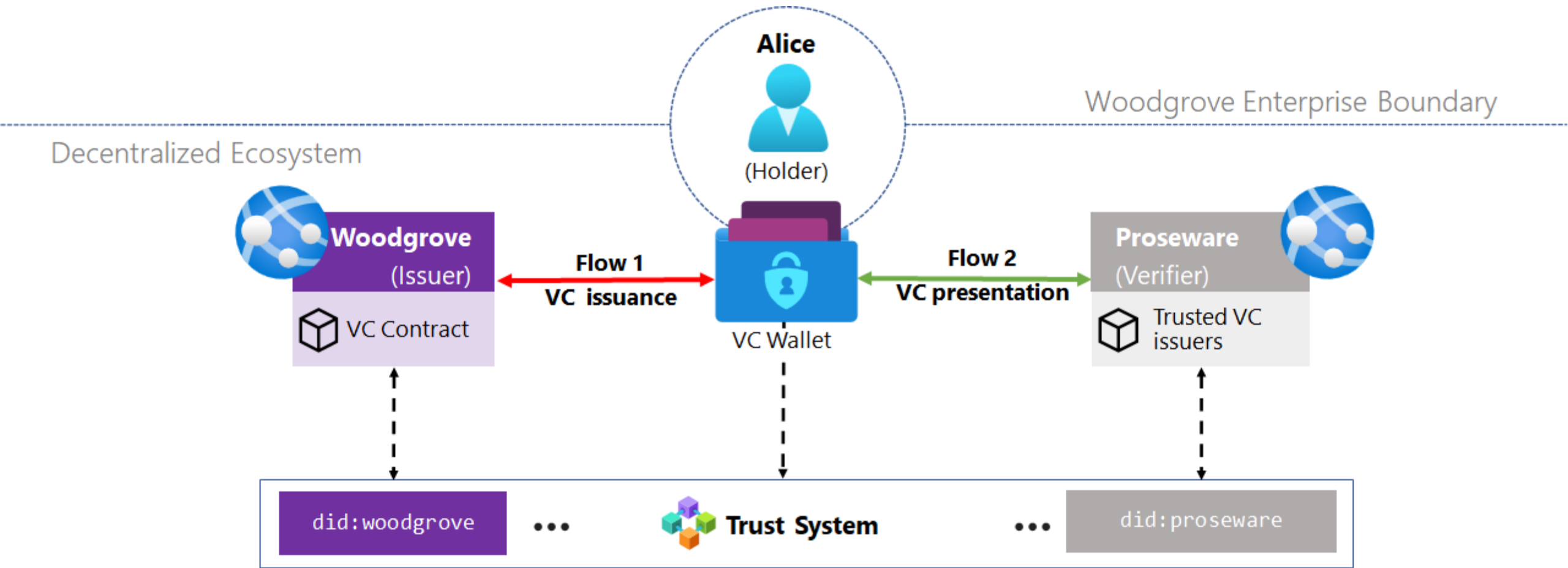
- Verifiable credentials are data objects consisting of claims made by the issuer attesting information about a subject.
- The schema identifies these claims.
- Claims include the DID of the issuer and the subject.
- The issuer's DID produces a digital signature as evidence of their attestation to this information.



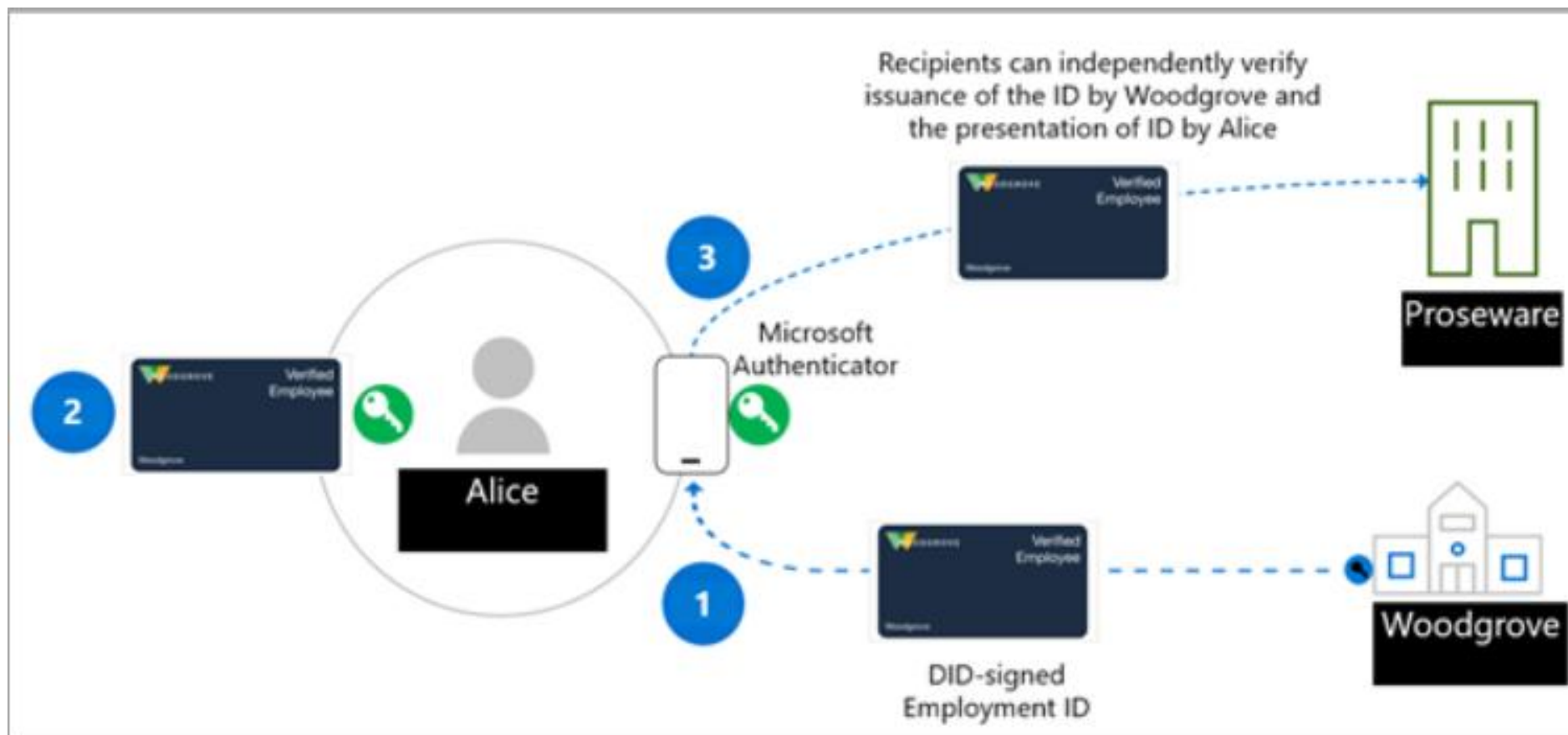
# Decentralized identity system



# Decentralized identity system



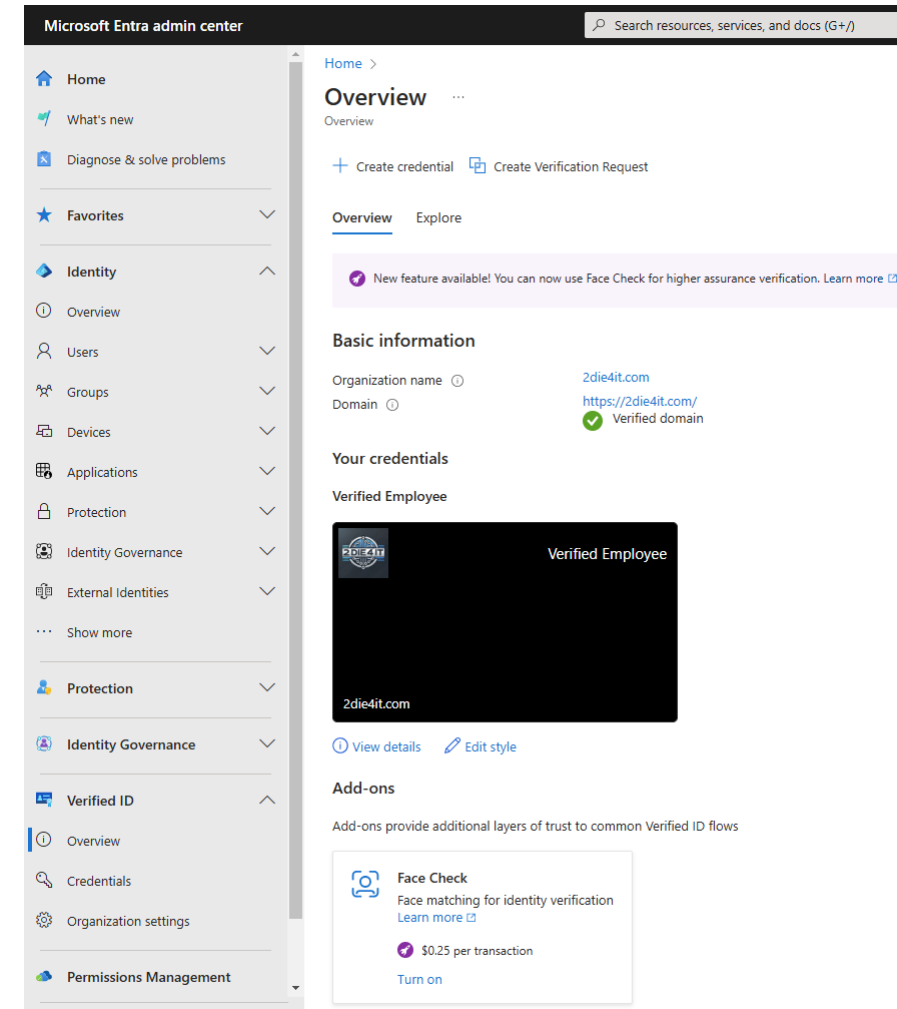
# Decentralized identity system





# How to start

- Microsoft Entra ID tenant
- Minimal license: Microsoft Entra ID Free (!)
- Enable Verified ID
  - Quick setup
  - Advanced setup
- Default VC: Verified Employee



# How to start: Quick setup (custom domain)



**Microsoft Entra admin center**

Search resources, services, and docs (G+ /)

Copilot

Home

What's new

Diagnose & solve problems

Favorites

Identity

Protection

Identity Governance

Verified ID

Overview

Credentials

Organization settings

Home >

Overview

Overview

Setup

Overview

## Welcome to Entra Verified ID

Verified ID enables fast remote onboarding, more secure access, and easy account recovery with a standards-based solution for individuals and organizations. [Learn more](#)

### Quick setup

Get ready in less than one minute with quick setup.

[Learn more](#)

Get started

### Advanced setup

Use advanced setup if you wish to set up manually.

[Learn more](#)

Advanced setup

## Unlock the power of Verified ID

Some examples of using Verified ID for improved business processes and enhanced security.

Microsoft Entra Verified ID

# How to start: Advanced setup



Microsoft Entra admin cent...

Search resources, services, and docs (G+/)

Home

Favorites

Identity

Protection

Identity governance

Verifiable credentials

Setup

Overview

Credentials

Organization settings

Permissions Management

Learn & support

Home > Setup >

Welcome to Entra Verified ID

Verified ID enables fast remote onboarding, more secure access, and easy account recovery with a standards-based solution for individuals and organizations. [Learn more](#)

1

Define organization settings

Confirm your organization's details, key vault, and Azure subscription information to create your organization's Decentralized Identity (DID).

Update

2

Register decentralized ID

Register your DID by uploading your unique DID document that'll allow applications and services to verify your credentials and for you to verify credentials from others.

Update

3

Verify domain ownership

To verify domain ownership, you'll need to upload a unique document that links your DID to the domain you entered in the organization settings.

Verify

Give feedback

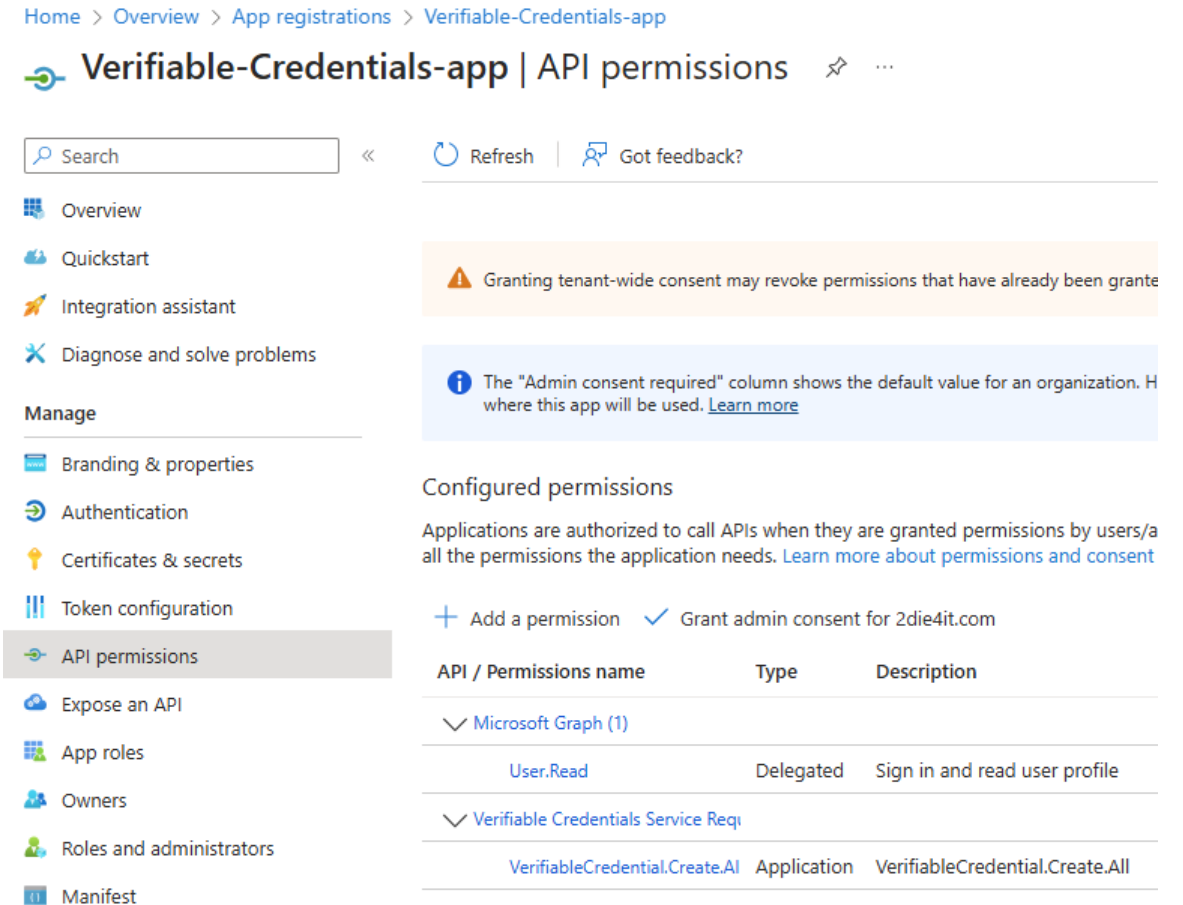
[Help improve this page](#)

Microsoft Entra Verified ID

# How to start: Advanced setup

More control about:

- Decentralized identifier (DID) name
- Own signing key (Azure Key Vault)
- App registration in Microsoft Entra ID



The screenshot shows the 'API permissions' page for the 'Verifiable-Credentials-app' in the Microsoft Entra ID portal. The left sidebar contains navigation links: Overview, Quickstart, Integration assistant, Diagnose and solve problems, Manage (Branding & properties, Authentication, Certificates & secrets, Token configuration, API permissions, Expose an API, App roles, Owners, Roles and administrators, Manifest), and API permissions (selected). The main content area has a search bar, a refresh button, and a 'Got feedback?' link. Below these are two informational banners: one warning that granting tenant-wide consent may revoke permissions, and another explaining the 'Admin consent required' column. The 'Configured permissions' section shows a table with one permission: 'User.Read' under the 'Microsoft Graph (1)' group. The table has columns for 'API / Permissions name', 'Type', and 'Description'. The 'User.Read' permission is of type 'Delegated' and has the description 'Sign in and read user profile'. Below this, there is a section for 'Verifiable Credentials Service Request' with a permission 'VerifiableCredential.Create.All' of type 'Application' and description 'VerifiableCredential.Create.All'.

Home > Overview > App registrations > Verifiable-Credentials-app

Verifiable-Credentials-app | API permissions

Search << Refresh Got feedback?

Overview

Quickstart

Integration assistant

Diagnose and solve problems

Manage

Branding & properties

Authentication

Certificates & secrets

Token configuration

API permissions

Expose an API

App roles

Owners

Roles and administrators

Manifest

Granting tenant-wide consent may revoke permissions that have already been granted

The "Admin consent required" column shows the default value for an organization. Here where this app will be used. [Learn more](#)

Configured permissions

Applications are authorized to call APIs when they are granted permissions by users/all the permissions the application needs. [Learn more about permissions and consent](#)

+ Add a permission ✓ Grant admin consent for 2die4it.com

| API / Permissions name                 | Type        | Description                     |
|----------------------------------------|-------------|---------------------------------|
| Microsoft Graph (1)                    |             |                                 |
| User.Read                              | Delegated   | Sign in and read user profile   |
| Verifiable Credentials Service Request |             |                                 |
| VerifiableCredential.Create.All        | Application | VerifiableCredential.Create.All |



# Default VC: Verified Employee

The screenshot displays the Microsoft Entra Verified ID console interface. On the left is a navigation pane with the following items: Home, Diagnose & solve problems, Favorites, Identity, Protection, Identity governance, Verified ID (expanded), Overview (selected), Credentials, Organization settings, Permissions Management, and Global Secure Access (Preview). The main content area is titled 'Overview' and includes a breadcrumb 'Home > Overview'. Below the title are tabs for 'Overview' and 'Explore'. A message states: 'Your verified employee credential is now ready. Let's try it out!'. Three numbered steps are provided: 1. Get your new credential (Users can obtain their verified employee credential by default through the MyAccount site. Link: Try it now), 2. Use your new credential (Test your new verified employee credential on a demo page to see how users share it when requested by a verifier. Link: Try it now), and 3. Control who can get the credential (All users in your tenant can obtain a verified employee credential by default. You can manage user eligibility. Link: Try it now). Below these steps is a 'Basic information' section showing 'Organization name' as 'Contoso' and 'Domain' as 'https://contoso.com/' with a 'Verified domain' status. The 'Your credentials' section shows 'Verified Employee' with a visual representation of the credential card. The card is black with a purple header, a green status indicator, and the text 'Verified Employee' and 'Contoso'. Below the card are links for 'View details' and 'Edit style'.

Home > Overview

Overview

Overview Explore

★ Your verified employee credential is now ready. Let's try it out!

- 1. Get your new credential**  
Users can obtain their verified employee credential by default through the MyAccount site.  
[Try it now](#)
- 2. Use your new credential**  
Test your new verified employee credential on a demo page to see how users share it when requested by a verifier.  
[Try it now](#)
- 3. Control who can get the credential**  
All users in your tenant can obtain a verified employee credential by default. You can manage user eligibility.  
[Try it now](#)

**Basic information**

Organization name ⓘ Contoso

Domain ⓘ <https://contoso.com/>  
✓ Verified domain

**Your credentials**

Verified Employee

Verified Employee

Contoso

ⓘ View details Edit style

# Default VC: Verified Employee



The screenshot shows the 'My Account' page for a verified employee. The header includes the 'WOODGROVE' logo and a 'My Account' dropdown. The left sidebar lists navigation options: Overview, Security info, Devices, Password, Organizations, Settings & Privacy, My sign-ins, Office apps, Subscriptions, and Give feedback. The main content area is divided into four sections. The top-left section displays the email 'mdunn@woodgrove.ms' and the location 'Tokyo', with a red circle highlighting the 'Get my Verified ID' link. The top-right section has an 'UPDATE INFO' link. The bottom-left section is titled 'Password' and includes a key icon and instructions to make the password stronger. The bottom-right section is titled 'Organizations' and includes a briefcase icon and instructions to see all organizations. The 'Devices' section on the right includes a laptop icon and instructions to disable lost devices.

WOODGROVE | My Account ▾

**Overview**

- Security info
- Devices
- Password
- Organizations
- Settings & Privacy
- My sign-ins
- Office apps
- Subscriptions
- Give feedback

mdunn@woodgrove.ms

Tokyo

[Get my Verified ID](#)

[Why can't I edit?](#)

[Sign out everywhere](#)

[UPDATE INFO >](#)

**Devices**

Disable a lost device and reconnected devices

[MANAGE DEVICES](#)

**Password**

Make your password stronger, or change it if someone else knows it.

**Organizations**

See all the organizations that you are a member of



# Default VC: Verified Employee

- Scan QR code and/or open link

Scan de QR-code met Microsoft Authenticator om uw geverifieerde id op te halen en op te slaan.



Deze QR-code is 5 minuten geldig. [Een nieuwe genereren](#)

[Kan de afbeelding niet worden gescand?](#)

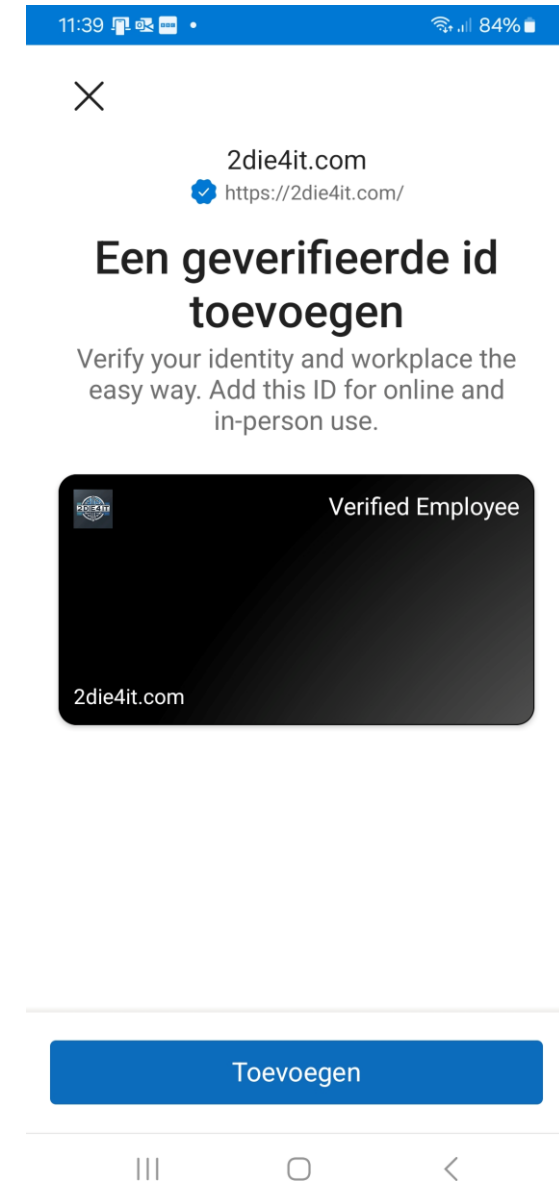
[Microsoft Authenticator downloaden](#)

Zodra de referentie is uitgegeven en opgeslagen in Authenticator, kunt u dit dialoogvenster sluiten.

**Gereed**

# Default VC: Verified Employee

- Add Verifiable Credentials to Microsoft Authenticator app



# Place of work verification on LinkedIn





**Pim Jacobs**  • 1st

Microsoft MVP & Principal Consultant  
Boxmeer

 Experience: InSpark | Innovate to Accelerate, InSpark, and 3 more

Pim has verifications

**Verifications** 

**Pim Jacobs**

**Workplace**  
InSpark | Innovate to Accelerate: Verified using Microsoft Entra Verified ID  
Over 1 year ago

[Learn how members verify information](#)

**Verify for free**  
Show authenticity to boost trust.

**Verify now**





# Place of work verification on LinkedIn



Learn / Microsoft Entra / Microsoft Entra Verified ID /

## Setting up place of work verification on LinkedIn (Preview)

Article • 12/13/2024 • 5 contributors [Feedback](#)

### In this article

- Deploying custom Webapp
- Architecture overview
- Frequently asked questions

**Important**

Place of work verification on LinkedIn is in public preview. Currently, customers with 10K seats or more are prioritized.



# Place of work verification on LinkedIn



Status update (7-2-2025)

We greatly appreciate your continued support and interest in Microsoft Entra Verified ID. We are in the process of making updates to enhance the Microsoft Entra Verified ID based employment history verification feature on LinkedIn. Upon completion, we will provide opportunities to test this functionality. While these improvements are in process, only email-based work verification will be available from the LinkedIn company page.

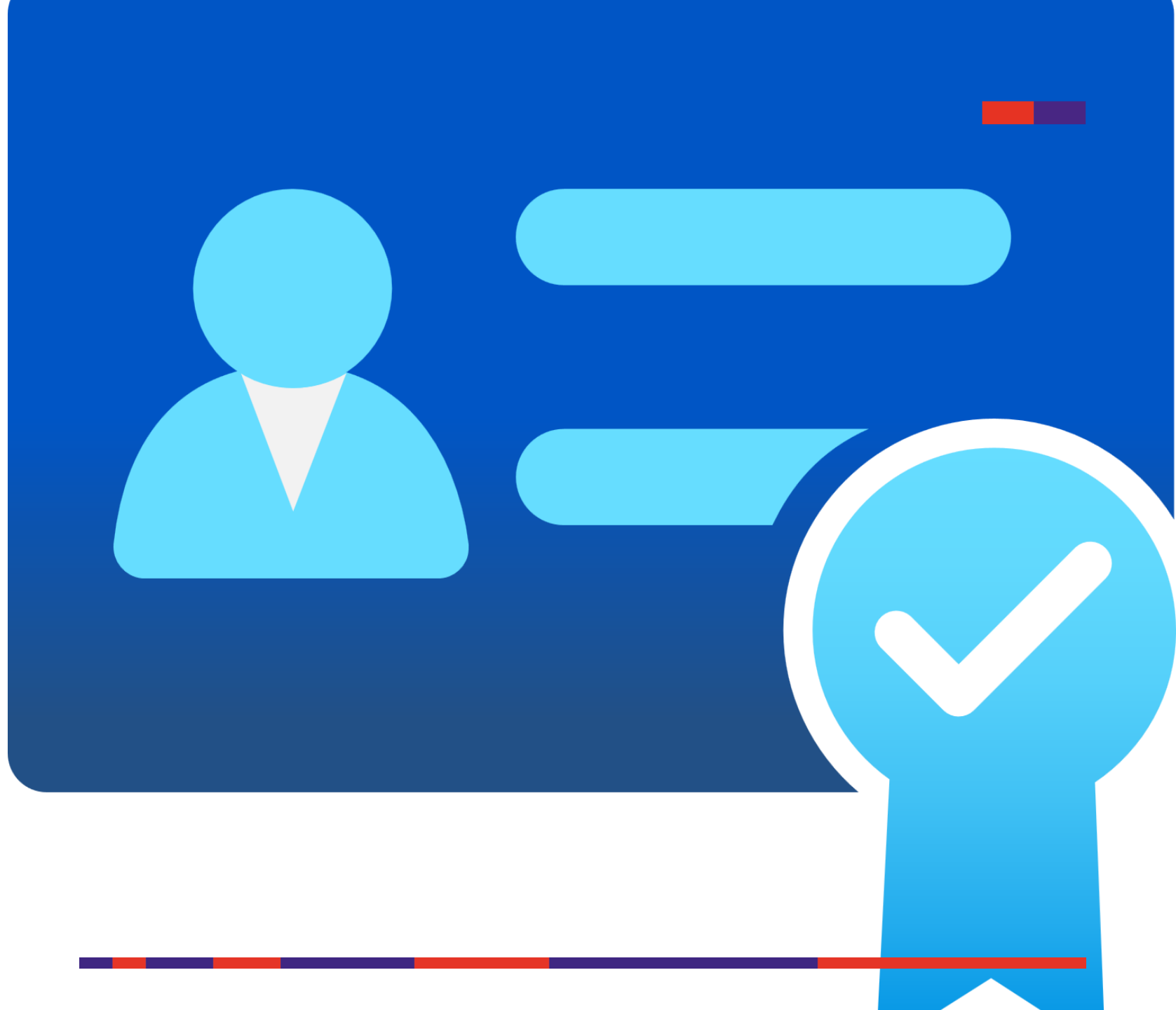
Thank you for your patience and support,  
The Microsoft Entra Verified ID team

[OwnYourIdentity@microsoft.com](mailto:OwnYourIdentity@microsoft.com)



# What's new

- Entitlement Management
  - Wallet Library
  - Validation
  - MyAccount available
  - Quick setup
  - Claims constraints
  - Face Check
  - Partner Gallery
  - New developments ?!
- Last: November 2025



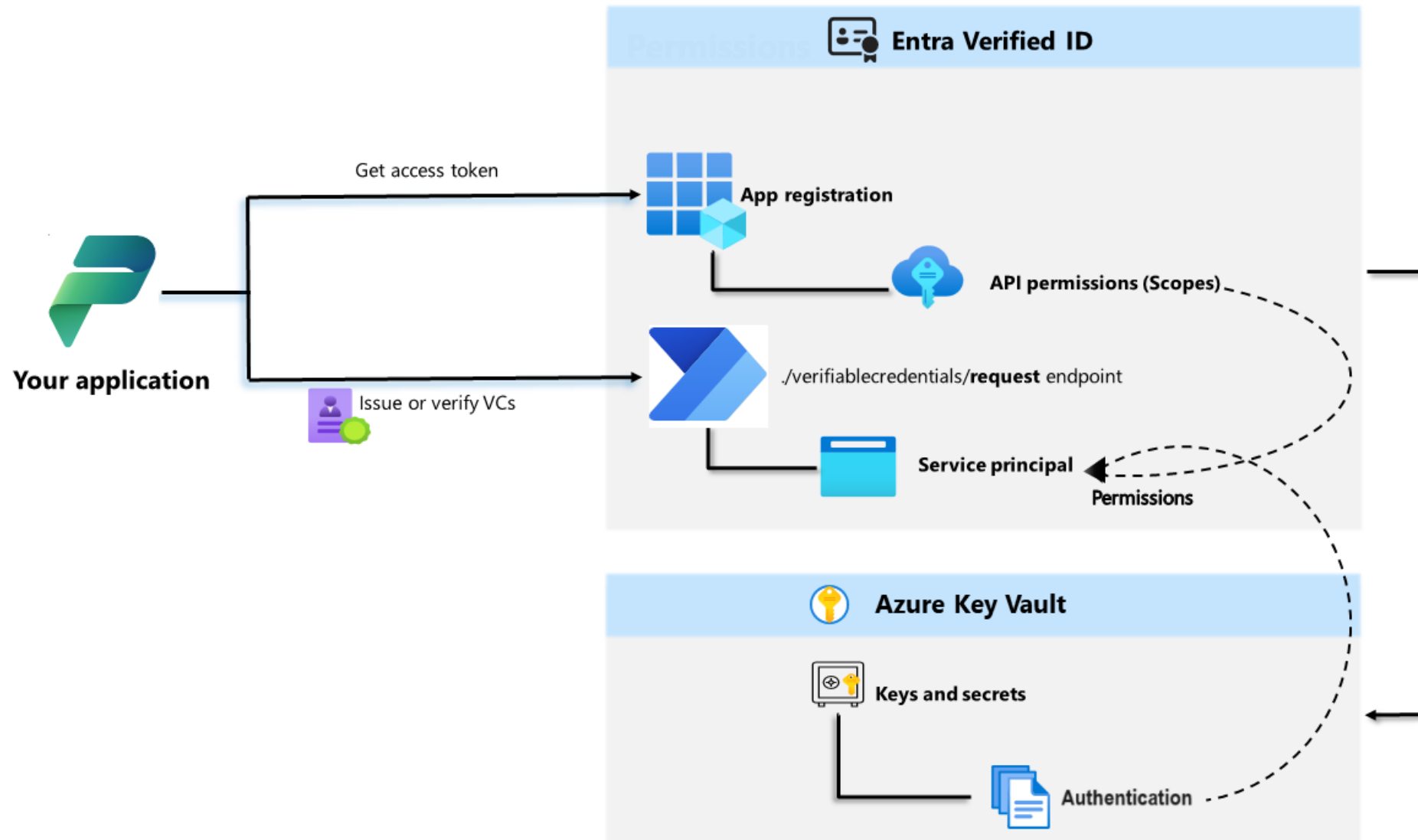
# Entra Verified ID: Use cases



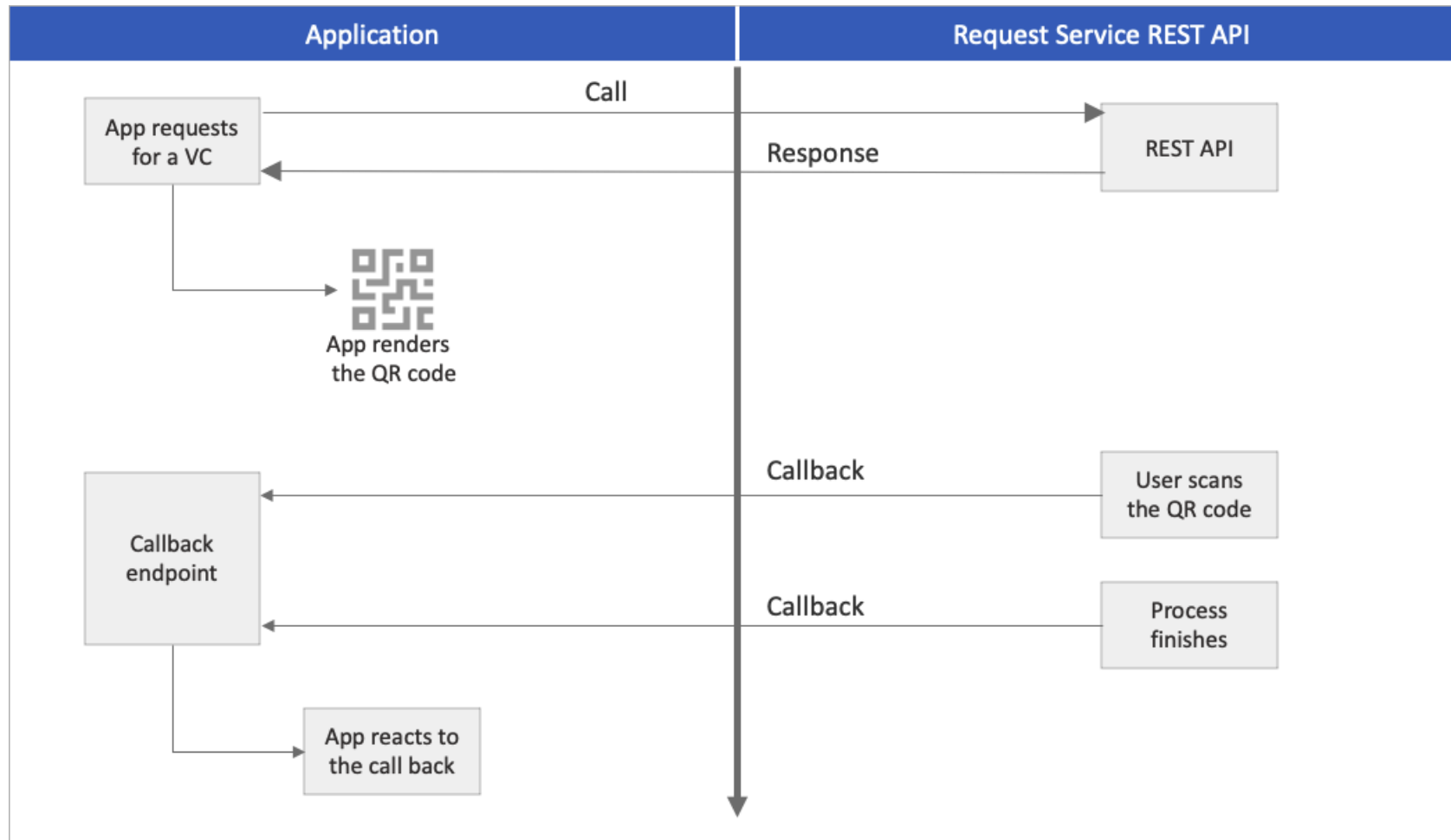
1. Verifying the identity of individuals from your organization or external partners for secure access and collaboration.
2. Issuing and verifying digital certifications, such as educational qualifications or safety credentials.
3. Using Verified Credentials as pick-up receipts for orders, including the option to delegate authorization to a colleague for added flexibility.
4. Providing participants of workshops or courses with a secure, verifiable credential as proof of attendance.



# Entra Verified ID with Power Platform



# Callback events





# Demo time !

# Entra Verified ID: Create Verifiable Credentials

| Create Verifiable Credentials | Claims                                                          |
|-------------------------------|-----------------------------------------------------------------|
| Directory based - Employee    | VerifiedEmployee: claims sourced from their user profiles       |
| Id token hint attestation     | Claim values passed in the issuance request payload             |
| Id token attestation          | Claims from the ID token of OpenID Connect identity provider    |
| Presentation attestation      | Claim values are taken from the presented verifiable credential |
| Self issued attestation       | You're required to manually enter values for the claims         |
| Multiple attestations         | Combination of attestations                                     |

# Entra Verified ID: Good option to secure issuance



Id token hint attestation + PIN code

- Just-in-time delivery of PIN code after scanning QR code

```
{
  "attestations": {
    "idTokenHints": [
      {
        "mapping": [
          {
            "outputClaim": "name",
            "required": true,
            "inputClaim": "$.name",
            "indexed": false
          },
          {
            "outputClaim": "email",
            "required": true,
            "inputClaim": "$.email",
            "indexed": true
          },
          {
            "outputClaim": "certification",
            "required": true,
            "inputClaim": "$.certification",
            "indexed": false
          },
          {
            "outputClaim": "date",
            "required": true,
            "inputClaim": "$.date",
            "indexed": false
          }
        ],
        "required": false
      }
    ],
    "validityInterval": 2592000,
    "vc": {
      "type": [
        "Certifications"
      ]
    }
  }
}
```



# Entra Verified ID: Better option to secure issuance



## Multiple attestations

- Presentation attestation + Id token hint
- Claim values are taken from the presented verifiable credential

```
{
  "attestations": {
    "idTokenHints": [
      {
        "mapping": [
          {
            "outputClaim": "userPrincipalName",
            "required": true,
            "inputClaim": "$.userPrincipalName",
            "indexed": false
          },
          {
            "outputClaim": "displayName",
            "required": true,
            "inputClaim": "$.displayName",
            "indexed": false
          },
          {
            "outputClaim": "mail",
            "required": true,
            "inputClaim": "$.mail",
            "indexed": false
          }
        ],
        "required": true
      }
    ],
    "presentations": [
      {
        "contracts": [
          "https://verifiedid.did.msidentity.com/v1.0/tenants/bb2c4d7b-8a1-4f1e-8b1e-1f1e1f1e1f1e"
        ],
        "credentialType": "VerifiedEmployee",
        "mapping": [
          {
            "outputClaim": "vcUserPrincipalName",
            "required": true,
            "inputClaim": "$.vc.credentialSubject.revocationId",
            "indexed": true
          },
          {
            "outputClaim": "vcDisplayName",
            "required": true,
            "inputClaim": "$.vc.credentialSubject.displayName",
            "indexed": false
          }
        ]
      }
    ]
  }
}
```

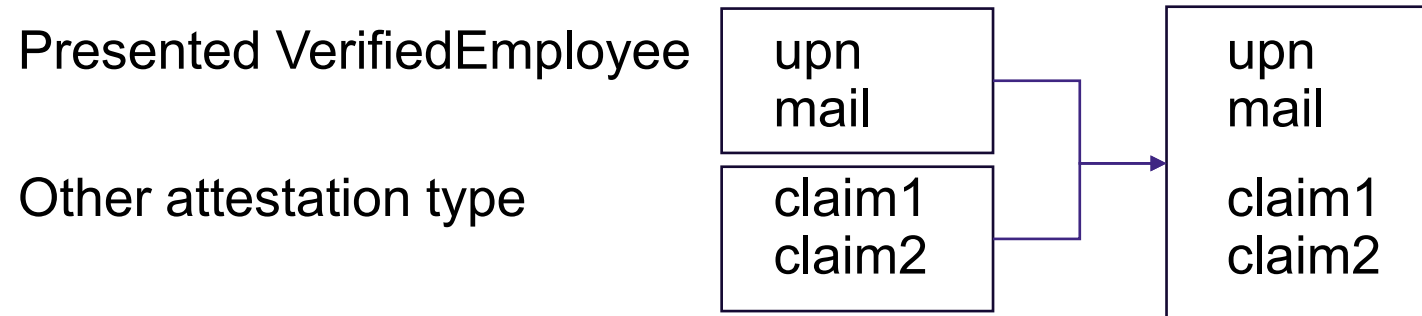


# Entra Verified ID: Better option to secure issuance



How to secure the issuance process

- Combine Presentation attestation (VerifiedEmployee) with other attestation type
  - Presentation attestation: claim values are taken from the presented VerifiedEmployee
  - Check claim values (upn, mail) in callback event  $\neq$  decline & revoke presented VC



# Entra Verified ID: Best option to secure issuance



## Id token attestation

- Claims from the ID token of OpenID Connect identity provider

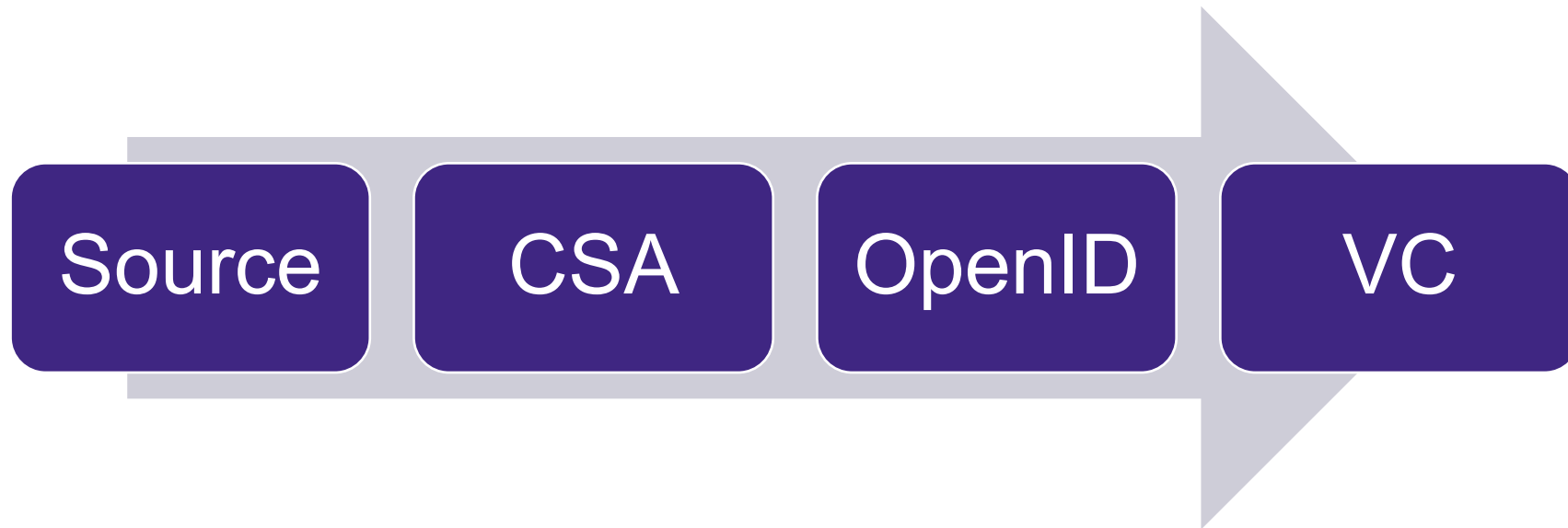
```
{
  "attestations": {
    "idTokens": [
      {
        "clientId": "bbb3f2d0-638a-477f-b1e0-3ecc811aa50a",
        "configuration": "https://login.microsoftonline.com/bb2c4d7b-8a8c-488d-bd3b-5d170bc63154/v2.0/.well-known/openid-configuration",
        "redirectUri": "vcclient://openid/",
        "scope": "openid profile email",
        "mapping": [
          {
            "outputClaim": "userName",
            "required": true,
            "inputClaim": "$.upn",
            "indexed": true
          },
          {
            "outputClaim": "displayName",
            "required": true,
            "inputClaim": "$.name",
            "indexed": false
          },
          {
            "outputClaim": "firstName",
            "required": true,
            "inputClaim": "$.given_name",
            "indexed": false
          },
          {
            "outputClaim": "lastName",
            "required": true,
            "inputClaim": "$.family_name",
            "indexed": false
          },
          {
            "outputClaim": "yearOfBirth",
            "required": false,
            "inputClaim": "$.yearOfBirth",
            "indexed": false
          }
        ]
      }
    ]
  }
}
```

# Entra Verified ID: Best option to secure issuance



Solution:

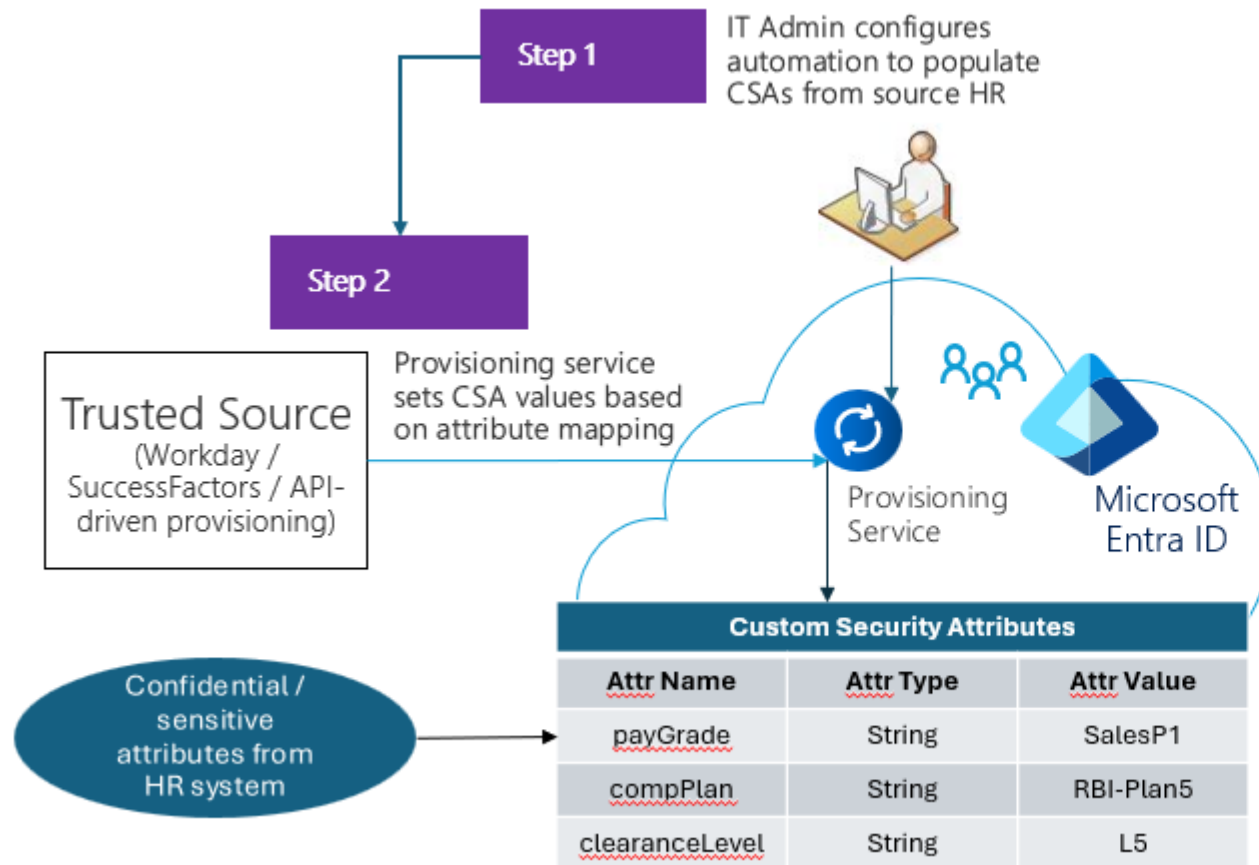
1. Custom security attribute provisioning using Microsoft Entra inbound provisioning capabilities
2. Issue Custom Security Attributes in Entra ID tokens
3. Issue custom Verifiable Credentials based on the OpenID Connect endpoint of the same app registration





# Entra Verified ID: Best option to secure issuance

## 1. Custom security attribute provisioning using Microsoft Entra inbound provisioning capabilities



Automatically sync sensitive HR attributes to Microsoft Entra ID as custom security attributes (CSA) using the provisioning service.

Supported sources:

- Workday
- SAP SuccessFactors
- Any HR system using API-driven provisioning

# Entra Verified ID: Best option to secure issuance



**Good Workaround!**

## 2. Issue Custom Security Attributes in Entra ID tokens

### Issuing Custom Security Attributes in Entra ID tokens

👤 Marius Solbakken 🕒 October 14, 2024

[Custom security attributes](#) in Entra ID is a feature that allows you to add attribute sets and attributes to your tenant, which you can use on all your user and application objects. A fairly common ask in Entra ID is the ability to store “secret data”, such as social security numbers, and making sure only certain principals can read these values. Here is where custom security attributes come into play. You can add an attribute set “PersonalInformation” to your tenant, and add “SSN” as an attribute to this attribute set. You can then tag each user with values, which only will be readable by those assigned the “Attribute assignment reader” role on tenant level or attribute set level.

However, currently Entra ID does not allow for these to be a part of issued tokens, not part of outbound synchronization rules and not as criteria for criteria based groups. So yeah, not really a first class citizen currently. For issued tokens, we can do something about it though!

The way we can address this, is to utilize the [Custom authentication extensions](#). We can essentially do this:

**Marius Solbakken**

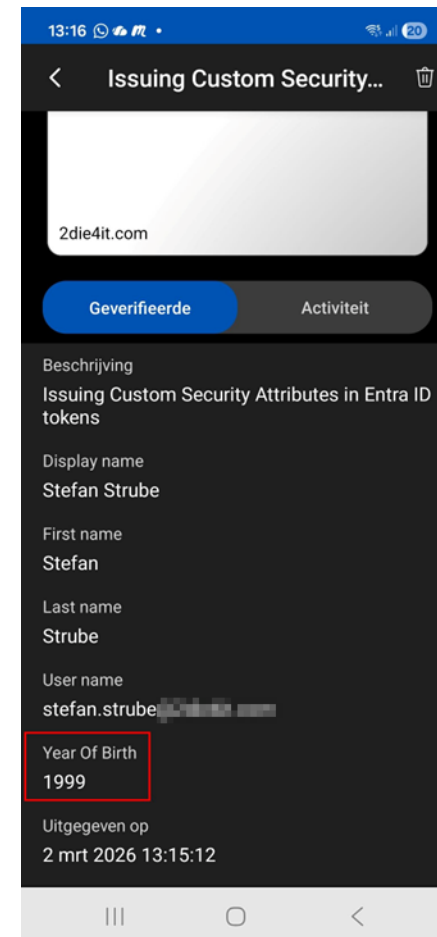
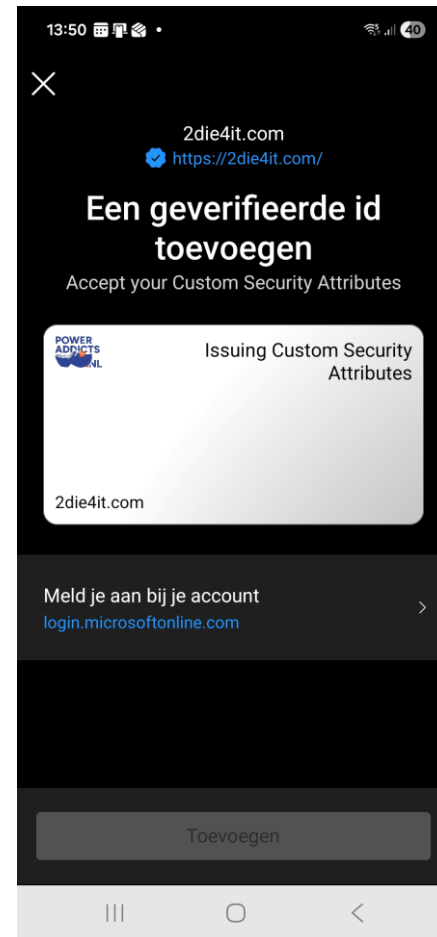
Co-founder and Principal Cloud Engineer @ [Fortytwo](#), Microsoft Security MVP, Host of the CloudFirst Podcast



# Entra Verified ID: Best option to secure issuance



3. Issue custom Verifiable Credentials based on the OpenID Connect endpoint of the same app registration



# Entra Verified ID: Security



How to secure the presentation process

- Constraints in createPresentationRequest
- Include Face Check as extra verification

How to secure the issuance process

- PIN code (ID token hint attestation only; bug with Android)
  - Actionable messages in Outlook
  - Adaptive cards in Teams, or SMS
- Id token attestation: claims from the ID token of OpenID Connect identity provider (authenticate)
  - Example: Custom Security Attributes from Entra ID



# Entra Verified ID: Create a revocable VC



- Microsoft Entra Verified ID doesn't store verifiable credential data
- The issuer needs to index one claim to make the credential searchable
- Only one claim can be indexed, and if there's none, you can't revoke credentials
- The selected claim to index is then salted and hashed and isn't stored as its original value
- How to revoke a credential ([index claim](#))
  - Via UI
  - Via Admin API
    - POST
    - /v1.0/verifiableCredentials/authorities/:authorityId/contracts/:contractId/credentials/:credentialId/revoke



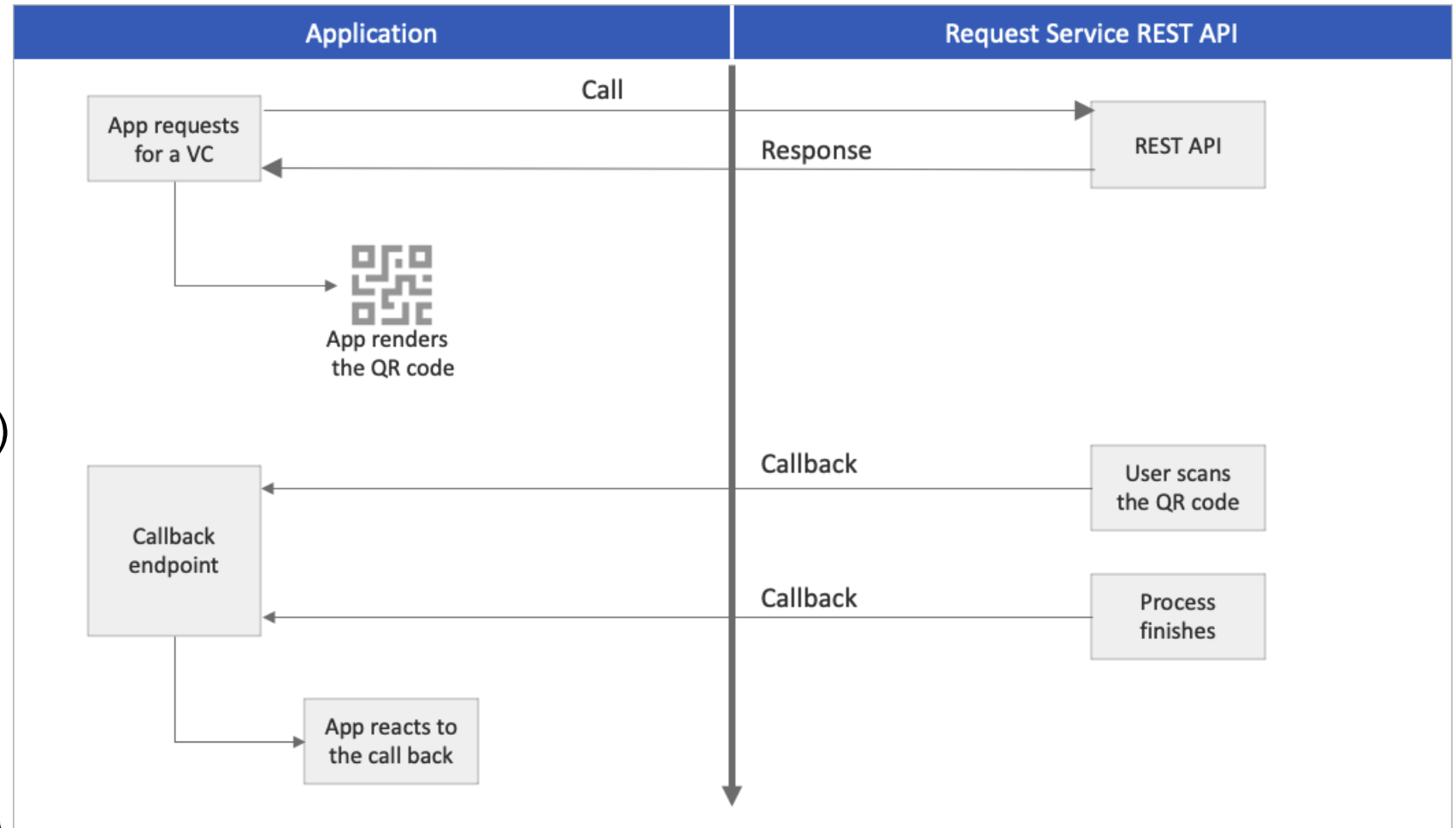
# Callback events

Request > Callback:

- url (https)
- state
- headers
  - api-key (API Key)
  - Authorization (Token)

Callback event:

- requestId
- requestStatus
- state
- verifiedCredentialsData
- subject



# Entra Verified ID: Advanced scenarios



- Using OpenID Connect (manifest)
  - Custom Security Attributes from Entra ID
- Using Face Check
- Using Verified ID with Entitlement Management
- Using the Wallet Library
- Integration with Azure AD B2C: B2C Custom Policies (Power Pages access)
- Integration with Copilot Studio
  
- Demo solutions
  - Onboard new remote employees
  - Verified helpdesk





# Entra Verified ID: Proof of attendance

